

Student ID:

Lab 05 Task

Your company, Make Abdullahpur Great Again (MAGA), is expanding its headquarters and requires a new network design to support various departments. The MAGA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix **192.(i).(i+3) % 10**. Below are the required devices for each of the departments:

$E(i) = (5i + 3) \bmod 41 + 5$	(Engineering)
$M(i) = (7i + 11) \bmod 26 + 5$	(Marketing)
$S(i) = (3i + 7) \bmod 36 + 5$	(Sales)
$A(i) = (11i + 17) \bmod 18 + 5$	(Administration)
$G(i) = (13i + 1) \bmod 9 + 5$	(Management)

where i is the last two digits your student ID

Implement and configure the topology, the devices should get their IP automatically from the DHCP server. Now for Engineering, Marketing and Sales the devices should use a separate DHCP server for automatic IP assignment & for Admin and Management they should use a router as a DHCP server. For each network you need to add minimum 4 end devices.

Use the same topology and configuration as the previous lab.

Now, your company is connected to the internet with another router called the Internet router. The number of global IP you have purchased is only 3, in which your own PC's local IP address has its own global IP address, the rest of the devices use the rest two IP addresses for accessing the internet. The global IP address starts from 110.i.i+1.0/24.

Lab 05 Grading Criteria

1. Correct Topology & Router Configuration (15 pts)

- Correct topology (5 pts)
- Correct router configuration (10 pts)

2. NAT Configuration (20 pts)

- Correct Static NAT Configuration (5 pts)
- Correct Non-Static NAT Configuration (10 pts)

3. Routing (5 pts)

4. Total: 35 pts